

Heavy-Hitter Masking for Sample-Efficient BabyLM Pretraining

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Abstract

We study whether bounded-memory heavy-hitter sketches can improve masked language model pretraining under BabyLM Strict-Small data constraints. Our method, Heavy-Hitter Masking (HHM), keeps the dataset, tokenizer, architecture, optimizer, training schedule, and total mask rate fixed, but reallocates part of the masked-token budget toward repeated local entity mentions and tokens that recently produced high prediction loss. Across three matched 10k-step local training runs, HHM does not improve validation loss over random masking: the random baseline reaches 3.0023 ± 0.1576 , while HHM reaches 3.0321 ± 0.1469 . Mask diagnostics show that the intervention is partially diluted by fallback random masking, with observed HHM masks split 15.4% entity, 18.9% error, and 65.6% random. Official BabyLM strict zero-shot evaluation gives a mixed counterpoint: HHM is slightly higher than the random baseline on the available BLiMP, Supplement, Entity Tracking, COMPS, and reading metrics, but the gains are small and sometimes high-variance. Follow-up pilots show that raw loss-guided CMS-Morph and Entity-Accuracy-Morph do not improve validation loss, while a simpler correctness-smoothed Accuracy-Morph variant improves matched validation loss in two seeds, with mean paired change -0.0155 ± 0.0051 , and raises Entity Tracking by $+5.51 \pm 3.27$. We therefore present HHM as a controlled mixed result and Accuracy-Morph as the most promising follow-up.

1 Introduction

The BabyLM challenge asks whether language models can learn robust linguistic generalizations from developmentally plausible data budgets rather than from web-scale corpora (Warstadt et al., 2023; Hu et al., 2024; Choshen et al., 2026). Under the Strict-Small track, models are trained on at most 10M words, making each prediction target valuable. Recent successful BabyLM systems often

improve efficiency by changing the model architecture, training objective, optimizer, or data pipeline (Charpentier and Samuel, 2024; Yu et al., 2024; Haller et al., 2025; Kosmopoulou et al., 2025). This paper instead asks how far a much smaller intervention can go.

We propose HEAVY-HITTER MASKING (HHM), a masking-only intervention for masked language model pretraining. Rather than changing the corpus or model, HHM changes which token positions are selected for prediction. The total mask rate remains fixed at 15%, matching the random masking baseline. This places the work in the line of adaptive MLM masking (Lee et al., 2022; Yang et al., 2023; Zhong et al., 2023; Edman and Fraser, 2025), but with a stricter ablation-style control: no external knowledge graph, no auxiliary loss, no subtoken embedding module, no architecture change, and no additional data.

The research question is: can bounded-memory heavy-hitter sketches improve small-data language model pretraining by reallocating a fixed masked-language-modeling budget toward repeated discourse entities and model-specific hard examples? HHM uses two compact signals: local entity heavy hitters, intended to emphasize recurring discourse participants, and online error heavy hitters, intended to revisit tokens the current model has recently struggled to predict.

The contribution is a controlled workshop-style evaluation of a cheap, inspectable masking policy. In the completed local training comparison, HHM is a small negative result on MLM validation loss: it reallocates the mask budget as intended, but its mean validation loss is slightly worse than random masking after 10k steps. This still clarifies the design space. The diagnostic logs show that entity and error heavy hitters account for only about one third of realized masks, with the rest filled by random fallback, suggesting that the current heuristic mixture may be too diluted to substantially improve

pretraining loss. Official BabyLM evaluation of the same checkpoints gives a more nuanced outcome: HHM is slightly higher on the available official zero-shot metrics, so the method may affect linguistic diagnostics even when it does not improve the local training objective.

We treat this negative result as part of the contribution rather than hiding it behind a single best run. The experiments separate three questions that are often bundled together in stronger systems: whether adaptive masking can be implemented without changing the training budget, whether raw model loss is a useful online difficulty signal, and whether subtoken difficulty can make the signal less brittle. This leads to the Accuracy-Morph follow-up, which replaces raw loss with smoothed top-1 correctness and adds character-trigram difficulty sketches while preserving the same 15% mask rate.

After the main comparison, we also tested masking-only follow-ups: CMS-Morph, Accuracy-Morph, and Entity-Accuracy-Morph. These pilots sharpen the current recommendation. Raw high-loss CMS sketches and entity-bonus decay do not improve validation loss, but the simpler Accuracy-Morph variant, which scores candidates by smoothed prediction correctness, improves matched validation loss in two seeds and produces the clearest Entity Tracking gains. We therefore frame HHM as a mixed controlled result and Accuracy-Morph as the most plausible next experiment rather than as a settled conclusion.

2 Background and Related Work

BabyLM. BabyLM evaluates language models trained under constrained data budgets, with particular emphasis on whether small models can acquire linguistic generalizations from limited input (Warstadt et al., 2023; Hu et al., 2024; Choshen et al., 2026). The 2026 Strict-Small track limits pretraining data to 10M words or less and asks participants to implement submissions using Hugging Face Transformers. The shared-task setting is useful for this paper because it makes the central constraint explicit: performance should improve by using limited data more effectively, not by silently increasing corpus size or model scale.

The recent BabyLM literature shows that strong results often come from changing several parts of the training recipe at once. The second BabyLM findings report that the best-performing submis-

sions tended to modify the training data, objective, or architecture, and that a hybrid causal-masked architecture was especially successful (Hu et al., 2024). GPT-BERT merges causal and masked objectives in a single transformer stack and outperforms masked-only or causal-only variants in the 2024 challenge setting (Charpentier and Samuel, 2024); AntLM similarly alternates causal and masked objectives and attention masks to combine the strengths of CLM and MLM (Yu et al., 2024). Other recent BabyLM systems explore orthogonal sources of efficiency, including linear or lightweight token mixers, optimizer changes, curated corpora, masked diffusion objectives, and frequency-informed training (Haller et al., 2025; Kosmopoulou et al., 2025). These systems are important points of comparison, but they answer a broader question than ours. We ask a narrower question: if the data, tokenizer, architecture, optimizer, schedule, training length, and total MLM mask rate are held fixed, can changing only the masking policy improve sample efficiency?

Why Mask Selection Matters. In masked language modeling, the model only receives a learning signal at the positions selected for prediction. The corpus may contain 10M words, but a 15% mask rate exposes the loss directly on only a fraction of token positions per epoch. Under web-scale pretraining this inefficiency may be tolerable because the model sees enormous redundancy. Under BabyLM Strict-Small constraints, however, the prediction budget is much tighter: repeatedly spending masks on very easy function words, isolated punctuation-adjacent tokens, or one-off examples may be less useful than spending some of the same budget on positions that stress lexical, discourse, or morphological generalization.

Adaptive Masking. Masked language modeling usually samples prediction targets uniformly from valid token positions. This is an attractive default because it is simple, unbiased with respect to token position, and does not require task-specific annotation. The downside is that random masking treats all valid positions as equally valuable prediction targets. Prior work has therefore explored masking policies that change what gets predicted. Concept-based curriculum masking evaluates token difficulty with linguistic and knowledge-graph criteria, then masks easy items first and related items later (Lee et al., 2022). Scheduled masking

changes the masking ratio and masked content over training time, arguing that time-invariant MLM settings are unlikely to be optimal (Yang et al., 2023). Self-Evolution learning focuses on informative or under-explored tokens and combines adaptive masking with token-specific label smoothing (Zhong et al., 2023). Most directly related to our follow-up experiments, Edman and Fraser (2025) adapt MLM masking probabilities in BabyLM according to the model’s ability to predict tokens and also use subtoken embeddings to improve morphological generalization.

HHM and Accuracy-Morph are in this adaptive-masking family, but the experimental emphasis is different. We keep the total mask rate fixed at 15% for every condition and avoid adding a new objective, label smoother, subtoken embedding module, external knowledge graph, or data-selection step. The only intended difference from the random baseline is the collator’s target-selection policy. This makes the experiment less likely to produce a large leaderboard gain, but it makes the causal interpretation cleaner: any change in validation loss or official diagnostics is attributable to which positions were predicted, not to a larger training signal.

Why Heavy Hitters. The first hypothesis behind HHM is that repeated strings in a local context are often linguistically important. In child-directed or narrative-like text, repeated names, pronouns, and content words help establish discourse participants and event structure. A random mask may hit these positions occasionally, but a heavy-hitter mechanism can deliberately spend part of the fixed mask budget on recurring entity-like mentions. This is meant to encourage the model to learn from local recurrence and to support entity-tracking behavior without adding a parser, a coreference system, or external supervision.

Why Hard Examples. The second hypothesis is curriculum-like: tokens that the current model repeatedly predicts poorly may be worth revisiting. The online error component of HHM records high-loss masked targets during training and later makes matching token strings more likely to be masked. This creates a lightweight feedback loop in which the model’s own errors influence future prediction targets. The risk is that raw loss can be noisy: rare words, tokenization artifacts, or inherently ambiguous contexts may receive attention without producing useful generalization. This risk motivated the

later Accuracy-Morph pilot, which replaces raw loss with smoothed prediction-correctness statistics.

Streaming Sketches. Streaming sketches are compact data structures that approximate properties of a stream, such as heavy hitters and frequency moments (Cormode and Muthukrishnan, 2005; Metwally et al., 2005). They are well matched to this setting because the masking policy observes a long stream of token events but should not store unbounded per-token histories. We use Space-Saving sketches to maintain bounded-memory summaries of repeated local entity strings and high-loss target tokens. The follow-up CMS-Morph pilots use Count-Min Sketches to track token and character-trigram difficulty signals. In both cases, the sketch is a small control mechanism over mask selection, not an added source of text or labels.

Morphological Motivation. The CMS-Morph and Accuracy-Morph pilots were tested because token-level difficulty may be too narrow in a small-data regime. If the model struggles with one word form, related forms may share prefixes, suffixes, stems, or orthographic patterns. Character trigrams are a deliberately simple proxy for these subtoken regularities: they require no morphological analyzer, work with the same tokenizer, and can generalize a difficulty signal across related surface forms. This connects to the BabyLM finding that subtoken information can improve morphological generalization (Edman and Fraser, 2025), but our implementation uses character-trigram sketches only to decide future mask positions. It does not change the input representation or add new embeddings.

Design Constraint. The central design constraint is that any observed difference should be attributable to mask selection rather than to extra data, a stronger model, a different tokenizer, or a larger prediction budget. This makes the experiment intentionally conservative. A positive result would indicate that a cheap online policy can make the same training budget more useful; a negative or mixed result still identifies which adaptive signals are too noisy or too weak under strict controls. This is also what makes the paper different from many successful BabyLM submissions: rather than proposing a stronger complete system, we isolate one low-cost mechanism and report both where it fails and where a corrected version begins to help.

3 Method

Overview. HEAVY-HITTER MASKING is implemented as a Hugging Face data collator for masked language modeling. It receives tokenized examples, selects valid non-special positions to mask, applies the same replacement policy as the random baseline, and returns metadata used to update the online error sketch. The method is intentionally placed in the collator rather than in the model: this makes the intervention cheap, easy to disable, and directly comparable to random masking. The collator is also the right level of abstraction for the scientific question. Prior adaptive masking work often improves a full training recipe by combining mask selection with schedules, auxiliary regularization, or representation changes (Yang et al., 2023; Zhong et al., 2023; Edman and Fraser, 2025); here the collator boundary enforces that the model and optimizer are unchanged.

Masking Mixture. HHM masks the same total proportion of tokens as the baseline: 15% of valid positions. The default mixture allocates 50% of masked positions to uniform random masking, 25% to local entity heavy hitters, and 25% to online error heavy hitters. The 50% random component is a guardrail: it keeps broad coverage over the corpus and prevents the adaptive policy from collapsing onto a narrow set of tokens. If either heavy-hitter source lacks enough positions, the remainder is filled by random valid positions. This fallback is part of the method rather than an implementation accident, because some batches simply do not contain enough repeated or previously hard tokens to support the target mixture.

Local Entity Heavy Hitters. For each sequence, a lightweight heuristic extracts entity-like token strings: alphabetic non-stopword tokens that are at least two characters long and either capitalized or content-word-like. A Space-Saving sketch with capacity 64 counts candidate strings. Positions belonging to repeated top entities are eligible for the entity-guided mask share. This component was tested because BabyLM evaluation includes entity-tracking diagnostics, and because repeated mentions are a place where a small model may benefit from seeing the same discourse participant under multiple contexts. The heuristic is deliberately shallow: the goal is not to solve entity recognition, but to ask whether a cheap recurrence signal is useful at all.

Online Error Heavy Hitters. After warmup, the training loop computes per-token MLM losses for masked positions. The top-loss masked tokens in each batch update a bounded Space-Saving sketch. Later batches preferentially mask positions whose token string appears in this error sketch. This tests whether the model can provide its own training signal: instead of assuming in advance which tokens matter, HHM lets recent prediction failures influence future masking. Warmup prevents the sketch from being dominated by the model’s earliest, largely uninformative errors. The negative HHM and CMS-Morph results show why this raw-loss signal is risky: it can overemphasize rare, ambiguous, or tokenization-sensitive items rather than stable learnable regularities.

CMS-Morph Pilot. CMS-Morph is an additional pilot variant of the same masking-only idea. It maintains two Count-Min Sketches: one over high-loss token strings and one over character trigram features extracted from high-loss tokens. The motivation is that a token-only sketch may memorize isolated hard words without transferring that signal to related forms. Character trigrams provide a small amount of subtoken generalization: if forms sharing a suffix or stem-like pattern are repeatedly hard, future examples with the same pattern can receive part of the mask budget. During training CMS-Morph keeps the total mask rate fixed at 15%, but after warmup allocates 70% of masked positions uniformly at random, 20% to token-sketch-guided positions, and 10% to trigram-sketch-guided positions. Sketches are updated only from masked-token losses in the training batches; validation and test data are never used for sketch updates.

Accuracy-Morph Pilot. Accuracy-Morph is a second pilot motivated by a failure mode of raw loss: high loss can reflect noise, ambiguity, or rare-token scale rather than a stable learnable difficulty. Accuracy-Morph therefore replaces raw high-loss updates with smoothed prediction-correctness statistics. For each masked token, the training loop records whether the model’s top prediction matches the MLM label. Two token sketches count masked-token exposures and wrong predictions, and two analogous sketches count character trigram exposures and wrong predictions. Candidate positions are scored by smoothed error rate,

$$\frac{\text{wrong} + 0.5}{\text{seen} + 1.0},$$

rather than raw loss, so a token or trigram becomes attractive only when it has been observed enough times and is often predicted incorrectly. The pilot uses a longer 30% warmup and a conservative 80/15/5 random/token/trigram mask mixture, again preserving the 15% total mask rate. This variant was tested to separate the idea of adaptive masking from the noisiness of raw-loss heavy hitters.

Entity-Accuracy-Morph Pilot. Entity-Accuracy-Morph tests whether the promising Accuracy-Morph signal can be improved by borrowing the repeated-token intuition from the strongest prior BabyLM systems. It uses the same correctness sketches and the same 30% warmup as Accuracy-Morph, but adds a small score bonus for repeated non-special token strings within a batch and restricts adaptive candidates to an intermediate predicted-error band. The band restriction was intended to avoid two extremes: tokens that are already too easy to be useful and tokens that may be too noisy or idiosyncratic to learn from. The adaptive share is also decayed late in training, moving from an 80/15/5 random/token/trigram mixture toward a 90% random mixture after 80% of training. This variant is intended to test whether entity-like repetition can help the correctness-smoothed masking signal without changing the model or total mask rate.

Design Principle. The method deliberately avoids changing the model, tokenizer, dataset, optimizer, schedule, or total mask rate. This keeps the comparison focused on the masking policy.

4 Experimental Setup

Data. We use the official detoxified BabyLM 2026 Strict-Small training set released by the BabyLM community. The dataset contains 10M words drawn from BNC spoken, CHILDES, Gutenberg, OpenSubtitles, Simple Wikipedia, and Switchboard sources. All compared systems use the same train/validation split.

Systems. The main comparison is between a standard random masked language modeling baseline and HHM. We additionally implement ablations for entity-only masking, error-only masking, a randomized sketch control, and a frequency-weighted masking control. After the main comparison, we also ran adaptive masking pilots: CMS-Morph, which replaces HHM’s entity/error sources with Count-Min Sketch scores over high-loss token

strings and high-loss character trigram features; Accuracy-Morph, which uses smoothed top-1 prediction correctness instead of raw loss as the adaptive signal; and Entity-Accuracy-Morph, which adds repeated-token scoring and late adaptive decay on top of Accuracy-Morph. CMS-Morph and Entity-Accuracy-Morph were run for seed 1; Accuracy-Morph was run for seeds 1 and 2 after the seed-1 pilot looked promising.

Why These Pilots. The follow-up pilots were chosen to test specific explanations for the HHM outcome rather than to tune toward the validation set. CMS-Morph asks whether the original token-level hard-example signal was too narrow and should generalize through subtoken patterns. Accuracy-Morph asks whether raw loss was too noisy and should be replaced by a more stable correctness rate. Entity-Accuracy-Morph asks whether the best pilot signal can be combined with repeated-token/entity pressure and a late return toward random masking. Each pilot keeps the same model and total mask rate, so the comparison remains about mask selection rather than capacity or compute.

The pilots also position the paper relative to prior adaptive masking work. Concept-based curriculum masking and scheduled masking show that changing what gets masked over training can improve MLM efficiency (Lee et al., 2022; Yang et al., 2023). BabyLM adaptive MLM work shows that model-predictability-aware masking and subtoken information can help in the strict-small setting (Edman and Fraser, 2025). Our pilots ask a smaller mechanistic question under tighter controls: which online signal is least noisy when the only allowed intervention is reweighting a fixed 15% mask budget? The answer in our runs is not raw cross-entropy loss, and not an entity bonus layered on top; the only replicated positive signal is smoothed top-1 correctness with a small character-trigram component.

Model and Training. All systems use the same Hugging Face Transformers masked language model, tokenizer, optimizer, batch size, sequence length, learning-rate schedule, random seed policy, training steps, and total mask rate. The only intended experimental difference is the masking policy. This differs from stronger BabyLM system submissions such as GPT-BERT, AntLM, BLaLM, or masked diffusion models, where the training

objective, architecture, optimizer, or model family changes (Charpentier and Samuel, 2024; Yu et al., 2024; Haller et al., 2025; Kosmopoulou et al., 2025). Those approaches are better candidates for maximizing leaderboard performance; our setup is better suited for isolating whether target selection alone carries useful signal.

Evaluation. We report validation MLM loss from the local training pipeline after 10k training steps. This is the closest measure to the training objective: the model is evaluated on held-out BabyLM-style text with the same masked language modeling loss, and lower loss indicates better average prediction of masked validation tokens. Validation loss is therefore the cleanest measure of sample-efficient pretraining under the objective we optimize, but it does not directly test whether the model has acquired specific linguistic generalizations.

We also evaluate checkpoints with the official BabyLM 2026 strict zero-shot pipeline for the tasks available in the local evaluation bundle. Table 1 summarizes the reported benchmarks. These evaluations use the trained masked language model without task-specific fine-tuning. For the accuracy-style benchmarks, the scorer compares the model’s preference over controlled alternatives, such as a grammatical versus ungrammatical sentence or a correct versus incorrect continuation. For reading, the scorer compares model-derived surprisal with human reading measurements and reports separate eye-tracking and self-paced reading scores. Higher is better for all official metrics.

BLiMP. BLiMP is a broad zero-shot linguistic acceptability suite built from minimal pairs. Each item contrasts a sentence that satisfies a targeted linguistic constraint with a minimally different sentence that violates it. The local report groups BLiMP items into fields such as morphology, semantics, syntax, and syntax/semantics, with individual phenomena including determiner-noun agreement, subject-verb agreement, binding, island constraints, negative polarity item licensing, passives, argument structure, and wh-dependencies. A model receives credit when it assigns the better score to the acceptable member of the pair.

BLiMP Supplement. The BLiMP Supplement extends the minimal-pair style evaluation to additional targeted phenomena not covered by the main BLiMP suite. In the local bundle, the reported

subsets include hypernymy, subject-auxiliary inversion, easy and tricky question-answer congruence, and turn taking. This metric is useful because it tests a mixture of syntactic, semantic, and discourse-pragmatic contrasts that may not be captured by the main BLiMP categories.

Entity Tracking. Entity Tracking is especially relevant to HHM because the method explicitly masks repeated entity-like strings. The benchmark tests whether a model can maintain information about discourse entities across controlled contexts. The local report includes regular cases, ambiguous-reference cases, and move-contents cases, each evaluated across increasing numbers of operations. Higher scores indicate that the model more often prefers the continuation or alternative consistent with the correct entity state. Because HHM was designed partly around discourse recurrence, Entity Tracking is the most directly motivated official diagnostic for the entity-heavy-hitter component.

COMPS. COMPS evaluates compositional generalization. The local report includes a base condition and nonce or “wug”-style conditions, including variants with distractors before or between relevant material. These examples test whether the model can combine lexical and structural information productively rather than relying only on memorized surface strings. This benchmark is relevant to the morph pilots because character-trigram adaptive masking is intended to encourage some generalization across related or novel-looking forms.

Reading. The reading benchmark evaluates whether the model’s word-by-word expectations align with human processing data. Instead of minimal-pair accuracy, the official report provides separate eye-tracking and self-paced reading scores. Intuitively, a better language model should assign higher surprisal to words that humans also find harder or slower to process, although the relationship is indirect. We treat these scores as psycholinguistic diagnostics rather than as direct measures of MLM training quality.

The main baseline and HHM checkpoints are evaluated across three seeds with both local validation loss and the available official metrics. The CMS-Morph and Entity-Accuracy-Morph pilots are evaluated with the same local and official commands for one seed, while Accuracy-Morph is evaluated for two matched seeds. We report these follow-ups separately from the main three-seed

Metric	Evaluation type	What it probes	Reported score
Validation MLM loss	In-domain held-out MLM	Fit to the held-out BabyLM-style validation split under the training objective	Mean cross-entropy; lower is better
BLiMP	Zero-shot minimal-pair preference	Morphology, syntax, syntax/semantics, negative polarity items, agreement, islands, binding, argument structure	Average accuracy; higher is better
BLiMP Supplement	Zero-shot minimal-pair preference	Additional targeted phenomena including hypernymy, question-answer congruence, subject-auxiliary inversion, and turn taking	Average accuracy; higher is better
Entity Tracking	Zero-shot controlled reference task	Whether the model tracks discourse entities through regular, ambiguous-reference, and move-contents conditions across increasing operation counts	Average accuracy; higher is better
COMPS	Zero-shot compositional generalization	Compositional interpretation, including nonce/wug-style conditions and distractor variants	Average accuracy; higher is better
Reading	Psycholinguistic prediction	Whether model surprisal aligns with human reading behavior in eye-tracking and self-paced reading data	Official reading scores; higher is better

Table 1: Evaluation metrics reported in this paper. The official BabyLM strict zero-shot pipeline scores the available benchmark files in the local evaluation bundle. Accuracy-style tasks compare model preferences over controlled alternatives; reading evaluates alignment between model-derived surprisal and human reading measurements.

comparison because they were selected adaptively after the initial HHM result. EWoK was not present in the local evaluation data, so it is not reported.

Diagnostics. Each run logs the number of total masked tokens and the number selected from entity, error, and random sources. This verifies that HHM reallocates the prediction budget without increasing it.

5 Results

Table 2 reports validation loss for the completed 10k-step main comparison, along with official BabyLM 2026 strict zero-shot evaluation results for the tasks available in the local evaluation bundle. EWoK was not present in the local evaluation data, so we omit it rather than substituting a non-identical task source.

Main Comparison. The main comparison is Baseline Random MLM versus HHM Entity+Error, with three 10k-step seeds each. The random baseline reaches a mean validation loss of 3.0023 ± 0.1576 , while HHM reaches 3.0321 ± 0.1469 , a $+0.0298$ increase in loss. On local MLM validation loss, HHM is therefore slightly worse than random masking in the completed matched-run comparison, although the gap is small relative to seed-to-seed variation.

The official zero-shot results point in the opposite direction, but only weakly. As summarized in Table 3, HHM is higher than the random baseline on every reported official metric: BLiMP improves by $+0.23$, BLiMP Supplement by $+0.11$, Entity Tracking by $+0.88$, COMPS by $+0.37$, eye-tracking reading score by $+0.31$, and self-paced reading score by $+0.50$. These are small gains, and Entity Tracking in particular has high HHM variance (± 4.39), so the official evaluation should be read as suggestive rather than decisive.

This pattern is important for interpretation. If we judged only by official averages, HHM would look mildly encouraging; if we judged only by validation loss, it would look negative. The controlled

design makes the tension visible. HHM changes the prediction targets but not the prediction budget, and the result is not a general sample-efficiency improvement. Instead, it appears to move some linguistic diagnostics without improving the objective used for pretraining. This is weaker than the gains reported by broader BabyLM systems that change architecture or objective, but it is a cleaner test of mask selection itself.

Significance Tests. Table 4 reports paired tests over the three matched seeds. None of the observed differences are statistically significant. The validation-loss gap against HHM has paired t -test $p = 0.114$ and exact sign-flip $p = 0.25$; COMPS and self-paced reading also have exact sign-flip $p = 0.25$ because all three paired deltas go in the same direction, but with only three seeds this is still the strongest possible two-sided exact result. We therefore interpret all main HHM differences as descriptive trends, not as statistically established effects.

No-Retraining Check. The official evaluation answers the main no-retraining question: the existing HHM checkpoints do show modest behavioral improvements despite slightly worse MLM validation loss. This weakens a purely negative interpretation of HHM, but does not establish a robust sample-efficiency gain. The bottom line is mixed: HHM is worse on the training objective used for selection, while the available official diagnostics show small average improvements.

Adaptive Masking Pilots. Table 5 reports three additional controlled seed-1 pilots. CMS-Morph keeps the same model, tokenizer, data, optimizer, schedule, training length, and 15% total mask rate, but after a 10% warmup it allocates 70% of masks randomly, 20% using a Count-Min Sketch over high-loss token strings, and 10% using a Count-Min Sketch over high-loss character trigrams. The realized final batches used the intended fixed budget: 1,216 masked tokens per batch, split into 832 random and 384 adaptive token/trigram-guided

System	Seeds	Val. Loss	BLiMP	Sup.	Entity	COMPS	Eye	SPR
Random MLM baseline	3	3.0023 $\pm$ 0.1576	62.43 $\pm$ 0.62	55.77 $\pm$ 0.16	16.77 $\pm$ 0.93	50.18 $\pm$ 0.30	1.97 $\pm$ 0.92	2.14 $\pm$ 0.38
HHM entity+error	3	3.0321 $\pm$ 0.1469	62.66 $\pm$ 0.51	55.88 $\pm$ 1.78	17.65 $\pm$ 4.39	50.55 $\pm$ 0.47	2.28 $\pm$ 0.14	2.64 $\pm$ 0.25

Table 2: Main results. Values are mean $\pm$ sample standard deviation over three 10k-step seeds. Validation loss is local MLM loss, where lower is better. Evaluation columns are official BabyLM strict zero-shot outputs available in the local evaluation bundle; higher is better. The local bundle did not include EWoK data, so EWoK is omitted.

Metric	Baseline	HHM	Δ
MLM validation loss	3.0023	3.0321	+0.0298
BLiMP	62.43	62.66	+0.23
BLiMP Supplement	55.77	55.88	+0.11
Entity Tracking	16.77	17.65	+0.88
COMPS	50.18	50.55	+0.37
Reading: eye tracking	1.97	2.28	+0.31
Reading: self-paced	2.14	2.64	+0.50

Table 3: Mean HHM change relative to the random masking baseline. Lower is better for MLM validation loss, so the positive loss delta is worse; higher is better for all official evaluation metrics.

masks.

CMS-Morph is not a clear improvement. Its final validation loss is 3.1574, slightly worse than the matched seed-1 random baseline at 3.1445 and slightly worse than HHM seed 1 at 3.1530. On official metrics, however, it is mixed rather than uniformly negative: relative to random seed 1, CMS-Morph improves BLiMP by +0.52, COMPS by +0.23, eye-tracking by +1.72, and self-paced reading by +0.52, while decreasing BLiMP Supplement by -1.36 and Entity Tracking by -0.03 . We therefore treat CMS-Morph as a useful additional probe, not as a replacement result: subtoken sketch guidance can move some official diagnostics, but this first mixture does not improve the core training objective.

Accuracy-Morph is more promising in the seed-1 pilot. It uses the same fixed 15% mask rate, but after a 30% warmup allocates 80% of masks randomly, 15% by token correctness sketches, and 5% by character trigram correctness sketches. The final batches again preserve the fixed budget, with 960 random and 256 adaptive masks per batch. Relative to random seed 1, Accuracy-Morph lowers validation loss by -0.0119 and improves Entity Tracking by +3.20, COMPS by +0.36, eye-tracking by +0.55, and self-paced reading by +0.33, while slightly decreasing BLiMP by -0.15 and BLiMP Supplement by -0.64 . This suggested that correctness-smoothed adaptive masking was a better next direction than raw loss-guided CMS-

Morph, motivating the seed-2 replication below.

The final limited-compute run replicates Accuracy-Morph with seed 2. As shown in Table 6, Accuracy-Morph seed 2 improves matched validation loss from 3.0296 to 3.0105, a -0.0191 change. It also improves Entity Tracking by +7.82, eye-tracking by +0.21, self-paced reading by +0.04, and BLiMP Supplement by +0.12, while leaving BLiMP unchanged and decreasing COMPS by -0.37 . This is the strongest current evidence for the masking idea: the raw-loss and entity-bonus variants remain negative or mixed, but correctness-smoothed masking now improves matched validation loss in an additional seed while preserving the fixed 15% mask budget.

Table 7 summarizes the matched two-seed comparison. Accuracy-Morph improves validation loss in both seed 1 and seed 2, with a mean paired validation-loss change of -0.0155 ± 0.0051 . Its largest diagnostic effect is Entity Tracking, where the mean paired gain is $+5.51 \pm 3.27$. The rest of the official metrics are mixed: BLiMP is nearly unchanged, Supplement is lower on average because of seed 1, COMPS is effectively flat, and both reading scores improve modestly. Table 8 gives the broader context, comparing the two Accuracy-Morph seeds against all three available random seeds.

The Accuracy-Morph result also clarifies what the paper can and cannot claim. It does not show that any adaptive mask is useful: CMS-Morph uses a larger adaptive share and still hurts validation loss. It also does not show that entity-like recurrence is sufficient: the entity-augmented variant loses the Accuracy-Morph advantage. The evidence instead points to a specific design lesson that matches the literature on model-aware masking: the adaptive signal should be stable and calibrated to the model’s actual prediction behavior. In this setting, smoothed correctness is more reliable than raw loss.

Entity-Accuracy-Morph tests a more targeted variant of Accuracy-Morph by adding repeated-token scoring, filtering adaptive candidates to an in-

Metric	Baseline mean	HHM mean	Δ HHM-Base	Paired t p	Exact sign-flip p
MLM validation loss	3.0023	3.0321	+0.0297	0.114	0.25
BLiMP	62.43	62.66	+0.23	0.757	1.00
BLiMP Supplement	55.77	55.88	+0.11	0.929	1.00
Entity Tracking	16.77	17.65	+0.88	0.721	0.75
COMPS	50.18	50.55	+0.38	0.177	0.25
Reading: eye tracking	1.97	2.28	+0.32	0.559	0.75
Reading: self-paced	2.14	2.64	+0.50	0.166	0.25

Table 4: Paired significance tests over the three matched random-vs-HHM seeds. Positive validation-loss deltas are worse because lower MLM loss is better; positive official-metric deltas are better. With only three paired seeds, the tests are intentionally low-power: the smallest possible two-sided exact sign-flip p value is 0.25.

System	Val. Loss	BLiMP	Sup.	Entity	COMPS	Eye	SPR
Random MLM, seed 1	3.1445	62.81	55.81	15.84	50.10	0.98	1.78
HHM, seed 1	3.1530	62.41	53.84	15.90	50.82	2.16	2.43
CMS-Morph, seed 1	3.1574	63.33	54.45	15.81	50.33	2.70	2.30
Accuracy-Morph, seed 1	3.1326	62.66	55.17	19.04	50.46	1.53	2.11
Entity-Accuracy-Morph, seed 1	3.1566	62.13	55.17	17.85	50.04	1.51	2.06
CMS-Morph - Random	+0.0129	+0.52	-1.36	-0.03	+0.23	+1.72	+0.52
Accuracy-Morph - Random	-0.0119	-0.15	-0.64	+3.20	+0.36	+0.55	+0.33
Entity-Accuracy-Morph - Random	+0.0121	-0.68	-0.64	+2.01	-0.06	+0.53	+0.28

Table 5: Additional one-seed adaptive masking pilots. CMS-Morph uses high-loss token and character trigram Count-Min Sketches; Accuracy-Morph instead uses smoothed prediction-correctness sketches and a more conservative adaptive share; Entity-Accuracy-Morph adds a repeated-token/entity bonus and late adaptive decay. All keep the total MLM mask rate fixed at 15%. Lower is better for validation loss; higher is better for official metrics.

intermediate predicted-error band, and decaying the adaptive share late in training. This did not improve the result. The best validation checkpoint reached 3.1342 at step 8,000, close to Accuracy-Morph, but the final validation loss rose to 3.1566. Relative to random seed 1, the official metrics improve Entity Tracking by +2.01, eye-tracking by +0.53, and self-paced reading by +0.28, but BLiMP falls by -0.68 and COMPS by -0.06 . The final batches preserved the fixed mask budget, with 1,152 random masks and only 64 adaptive masks after late decay. We therefore treat this run as evidence against adding entity bonuses and late adaptive decay in the current form; the cleaner Accuracy-Morph pilot remains the stronger follow-up.

Table 9 summarizes the current situation. The paper should not claim that HHM is a strong positive result: the three-seed validation-loss comparison is negative, while the official metrics provide only suggestive behavioral gains. Among the follow-up pilots, Accuracy-Morph is now the only branch with a replicated matched-seed validation-loss improvement and a strong entity-tracking diagnostic. The recommended continuation is therefore an additional Accuracy-Morph seed, not raw-loss or

entity-bonus variants.

Ablation Results. Table 10 reports single-seed preliminary ablations for entity-only masking, error-only masking, randomized sketch masking, and frequency-only masking. These runs were short pipeline checks rather than full comparisons: their validation losses remain near 8.65–8.72 after 20 steps, with the randomized sketch control lowest among them at 8.6462. They are useful for verifying implementation coverage but should not be interpreted as substantive ablation evidence. The current completed evidence is therefore the three-seed random-vs-HHM comparison plus the adaptive follow-up runs, not a full multi-seed ablation suite.

Mask Diagnostics. Table 11 summarizes the HHM mask-source logs. The observed HHM allocation is 15.4% entity-selected masks, 18.9% error-selected masks, and 65.6% random masks. This confirms that the collator is reallocating part of the fixed 15% mask budget toward heavy hitters, while also showing that fallback to random masking is common under the current heuristics. Table 12 reports the corresponding final-step budget checks for the adaptive pilots.

6 Masking Analysis

We analyze whether HHM changes the distribution of prediction targets while preserving the same total number of masked tokens as the baseline. The implementation logs total masked tokens, entity-selected masks, error-selected masks, and fallback random masks for every training step.

Fairness Checks. Unit tests verify that random masking and HHM mask the same number of valid positions, that padding and special tokens are never masked, that entity positions correspond to re-

System	Seed	Val. Loss	BLiMP	Sup.	Entity	COMPS	Eye	SPR
Random MLM	2	3.0296	62.76	55.91	17.70	50.51	2.12	2.53
Accuracy-Morph	2	3.0105	62.76	56.03	25.52	50.14	2.33	2.57
Accuracy-Morph – Random	2	-0.0191	+0.00	+0.12	+7.82	-0.37	+0.21	+0.04

Table 6: Final limited-compute experiment: Accuracy-Morph seed 2 compared with the matched random MLM seed 2 baseline. Lower is better for validation loss; higher is better for official metrics.

System	Seeds	Val. Loss	BLiMP	Sup.	Entity	COMPS	Eye	SPR
Random MLM	1,2	3.0871 $\pm$ 0.0812	62.79 $\pm$ 0.04	55.86 $\pm$ 0.07	16.77 $\pm$ 1.32	50.31 $\pm$ 0.29	1.55 $\pm$ 0.81	2.16 $\pm$ 0.53
Accuracy-Morph	1,2	3.0715 $\pm$ 0.0863	62.71 $\pm$ 0.07	55.60 $\pm$ 0.61	22.28 $\pm$ 4.58	50.30 $\pm$ 0.23	1.93 $\pm$ 0.57	2.34 $\pm$ 0.33
Paired delta	1,2	-0.0155 $\pm$ 0.0051	-0.08 $\pm$ 0.11	-0.26 $\pm$ 0.54	+5.51 $\pm$ 3.27	-0.00 $\pm$ 0.52	+0.38 $\pm$ 0.24	+0.19 $\pm$ 0.21

Table 7: Matched-seed Accuracy-Morph comparison. The paired delta is Accuracy-Morph minus Random MLM for seeds 1 and 2. Accuracy-Morph improves validation loss in both matched seeds and produces a large positive Entity Tracking delta, while BLiMP, Supplement, and COMPS remain roughly flat or slightly lower.

peated local strings, that the error sketch is inactive during warmup, and that masking is deterministic under a fixed seed.

Observed Allocation. Across the three completed 10k-step HHM runs, the collator masks the same average number of tokens per step as the random baseline: 1215.8 valid tokens. The source of those HHM masks is 15.4% entity-selected positions, 18.9% error-selected positions, and 65.6% fallback random positions. This verifies that HHM changes the target distribution without increasing the total prediction budget, but it also shows that the intended 25/25/50 entity-error-random mixture is not fully realized.

Interpretation. The high fallback rate gives a concrete explanation for the weak validation-loss result. Entity and error sketches are active, but the realized intervention is closer to a mostly random masking policy with a moderate heavy-hitter bias than to an evenly balanced three-source mixture. The official evaluation suggests that this moderate shift may still affect measured linguistic behavior: HHM is slightly higher than the random baseline on the available official zero-shot tasks, including Entity Tracking and COMPS. However, the gains are small and often well within seed variation, so they should be treated as a diagnostic signal rather than a reliable win.

The pilot diagnostics point to a narrower lesson. Increasing adaptivity is not automatically helpful: CMS-Morph assigns a larger final adaptive share than Accuracy-Morph but has worse validation loss, while Entity-Accuracy-Morph decays to a very small final adaptive share and also loses

the Accuracy-Morph advantage. The useful signal appears to be the scoring rule, not simply the amount of non-random masking. In the current runs, smoothed prediction correctness is a cleaner adaptive signal than raw loss or repeated-token bonuses.

7 Discussion

The central question is whether compact memory can make small-data pretraining more efficient without changing the model or data. HHM tests this by reallocating a fixed prediction budget toward repeated discourse entities and online hard examples.

The completed local validation results do not support HHM as an improvement over random masking in its current form. The gap is small, but it is in the wrong direction: HHM ends at 3.0321 mean validation loss versus 3.0023 for random masking. By the training objective, HHM is therefore a controlled negative result rather than evidence for sample-efficiency gains.

The negative validation result is still informative because the diagnostics expose how the intervention behaves. HHM does not violate the fairness constraint: it masks the same number of tokens as the baseline. However, the realized mask mixture is substantially more random than the target mixture because heavy-hitter sources often lack enough eligible positions. This suggests that the current entity heuristic, error-key granularity, or mixture weights may be too conservative to improve MLM loss.

The official evaluation complicates the story in a useful way. Without retraining or changing checkpoints, HHM is slightly higher than the random

System	Seeds	Val. Loss	BLiMP	Sup.	Entity	COMPS	Eye	SPR
Random MLM	1,2,3	3.0023 $\pm$ 0.1576	62.43 $\pm$ 0.62	55.77 $\pm$ 0.16	16.77 $\pm$ 0.93	50.18 $\pm$ 0.30	1.97 $\pm$ 0.92	2.14 $\pm$ 0.38
Accuracy-Morph	1,2	3.0715 $\pm$ 0.0863	62.71 $\pm$ 0.07	55.60 $\pm$ 0.61	22.28 $\pm$ 4.58	50.30 $\pm$ 0.23	1.93 $\pm$ 0.57	2.34 $\pm$ 0.33

Table 8: Aggregate view after the final Accuracy-Morph seed-2 run. Accuracy-Morph now has two seeds, while the random baseline has three. The aggregate is therefore useful as context but not a perfectly balanced significance test.

Branch	Evidence	Seeds	Current interpretation
HHM entity+error	Worse MLM loss than random (+0.0298), but small gains on all available official metrics	3	Mixed diagnostic result; not a clear training-objective improvement
CMS-Morph	Worse seed-1 MLM loss than random (+0.0129), mixed official metrics	1	Useful probe, but raw high-loss sketches are not the best next path
Accuracy-Morph	Better matched seed-2 MLM loss than random (-0.0191), unchanged BLiMP, and much higher Entity Tracking (+7.82)	2	Best current follow-up; positive but still needs a third seed
Entity-Accuracy-Morph	Worse seed-1 MLM loss than random (+0.0121), smaller Entity Tracking gain than Accuracy-Morph	1	Do not continue this variant as the main direction

Table 9: Current status of the experimental branches. Positive validation-loss deltas are worse because lower MLM loss is better; official metric deltas are interpreted only as diagnostics unless replicated across seeds.

Condition	Steps	Validation loss
Random baseline	20	8.6524
HHM entity+error	20	8.6703
Entity only	20	8.6535
Error only	20	8.6611
Random sketch control	20	8.6462
Frequency only	20	8.7238

Table 10: Preliminary single-seed component ablations. These are 20-step pipeline checks rather than full training runs, so they verify implementation behavior but are not used as substantive evidence about final model quality.

System	Entity	Error	Random
HHM entity+error	15.4%	18.9%	65.6%

Table 11: Observed source of masked positions in the HHM diagnostic logs, aggregated over 30,000 logged batches from the three 10k-step HHM seeds. The target mixture is 25% entity, 25% error, and 50% random, with random fallback used when a heavy-hitter source lacks enough valid positions.

baseline on each available official zero-shot metric: BLiMP, Supplement, Entity Tracking, COMPS, and both reading scores. These gains are not large enough to claim a robust win, especially given the entity-tracking variance, and the paired tests over three seeds do not show statistically significant differences. They nevertheless show that worse local MLM loss does not necessarily imply uniformly worse BabyLM behavior. The paper should therefore be read as a mixed result: bounded-sketch masking is mechanically viable and may nudge downstream linguistic diagnostics, but the current simple policy is not yet a convincing training-objective improvement.

The later adaptive masking pilots sharpen the next-step recommendation. Raw high-loss CMS-

Pilot	Final masks	Random masks	Adaptive masks
CMS-Morph	1,216	832	384
Accuracy-Morph	1,216	960	256
Entity-Accuracy-Morph	1,216	1,152	64

Table 12: Final-step mask-budget diagnostics for the seed-1 adaptive pilots. All pilots preserve the same 15% total MLM mask rate; they differ only in how much of that fixed budget is assigned to adaptive token or character-trigram sources. Entity-Accuracy-Morph decays to a much smaller adaptive share late in training.

Morph changes some official diagnostics but worsens validation loss. Accuracy-Morph, which scores candidates by smoothed prediction correctness, is the only follow-up that improves matched validation loss while also producing a large Entity Tracking gain. The seed-2 replication strengthens this conclusion: across seeds 1 and 2, Accuracy-Morph improves matched validation loss by -0.0155 ± 0.0051 and Entity Tracking by $+5.51 \pm 3.27$. It does not create a broad official-evaluation win, since Supplement is lower on average and COMPS is flat, but it is the only variant whose local objective and targeted diagnostic both move in the desired direction. The entity-augmented follow-up does not preserve this advantage, suggesting that repeated-token bonuses and late adaptive decay are not the right additions in their current form. If this line is continued, the most defensible path is to run a third Accuracy-Morph seed rather than to add more heuristic components.

Relative to the closest ACL Anthology work, the main difference is experimental scope. Prior adaptive-masking papers show that changing masked content, schedules, or predictability-aware token probabilities can improve pretraining or BabyLM performance (Lee et al., 2022; Yang et al., 2023; Edman and Fraser, 2025). BabyLM system papers show that much larger changes, such as hybrid causal-masked objectives, lightweight

architectures, curated data, optimizer changes, or masked diffusion, can be stronger overall routes to leaderboard performance (Hu et al., 2024; Charpentier and Samuel, 2024; Yu et al., 2024; Haller et al., 2025; Kosmopoulou et al., 2025). Our experiments are deliberately narrower. The value of the paper is not that HHM beats those systems; it does not. The value is that it isolates the masking-policy variable and documents which signals survive that constraint. Under this stricter test, raw loss and entity recurrence are not enough, while correctness-smoothed token and trigram sketches are a plausible low-cost direction.

8 Limitations

The current implementation uses heuristic token-level entity extraction rather than a full syntactic or coreference system. Error heavy hitters are keyed by target token string in the first implementation, so they do not distinguish different contexts for the same token. The approach may overemphasize repeated or difficult tokens without improving the linguistic generalizations measured by BabyLM.

The local training comparison and the available official BabyLM strict zero-shot evaluations are complete for the two main systems, but the local evaluation bundle did not include EWOK. We therefore do not report EWOK and cannot assess that part of the full BabyLM suite here. The diagnostic logs also show substantial fallback to random masking, so these results should not be interpreted as a strong test of a perfectly balanced entity/error/random target mixture. Finally, the official HHM gains are small and based on three seeds; paired significance tests do not establish reliable differences. They should be treated as suggestive diagnostics rather than a robust downstream improvement claim. CMS-Morph and Entity-Accuracy-Morph remain single-seed follow-ups, and Accuracy-Morph has only two matched seeds, so the follow-up results identify promising directions rather than final comparative performance. The component ablations reported here are short 20-step checks and should not be read as final ablation evidence.

The paper also does not attempt to match the strongest BabyLM systems. In particular, we do not use hybrid causal-masked objectives, architecture changes, optimizer changes, curated corpora, external lexical resources, or auxiliary representation modules. That restriction is scientifically useful for attribution, but it limits expected abso-

lute performance. A stronger final system would likely combine the best controlled masking signal here, Accuracy-Morph, with a stronger BabyLM backbone such as a hybrid causal-masked model; that experiment would answer a different question from the one isolated in this paper.

9 Ethics Statement

This work studies corpus selection for small language model training. The approach may affect which linguistic varieties and domains are represented in the selected corpus. We therefore report corpus statistics and encourage analysis of distributional changes introduced by selection.

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To be completed.

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A Implementation Details

The code is organized in `./babylm-hhm`. Core masking classes are in `src/masking/`. Configurations cover the random baseline, HHM, entity-only, error-only, randomized sketch control, frequency-only control, CMS-Morph, Accuracy-Morph, and Entity-Accuracy-Morph. The official BabyLM evaluation repository is available locally.

B Reproducibility Commands

Tests are run with the local test runner. The three-seed main comparison uses the experiment runner with the `main` suite and seeds 1–3. The preliminary ablations use the same runner with the `ablations` suite. The adaptive pilots use the training entry point with the CMS-Morph, Accuracy-Morph, and Entity-Accuracy-Morph config files. All full training runs were executed with CUDA visible.